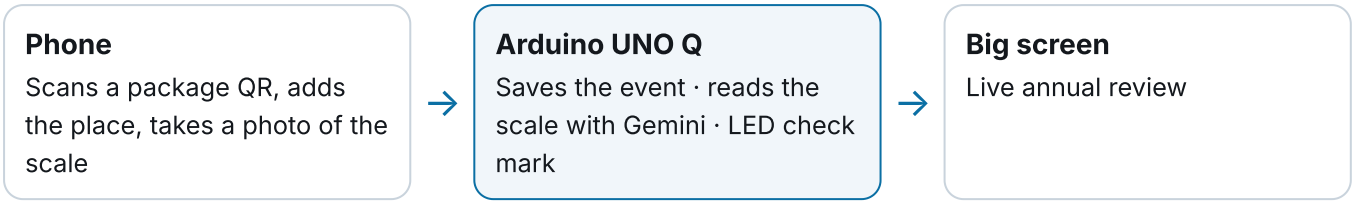
Why a hardware gateway?

Questions and answers about the Arduino UNO Q in ThreadTrace · Felix Liu · Since AI Hackathon 2026

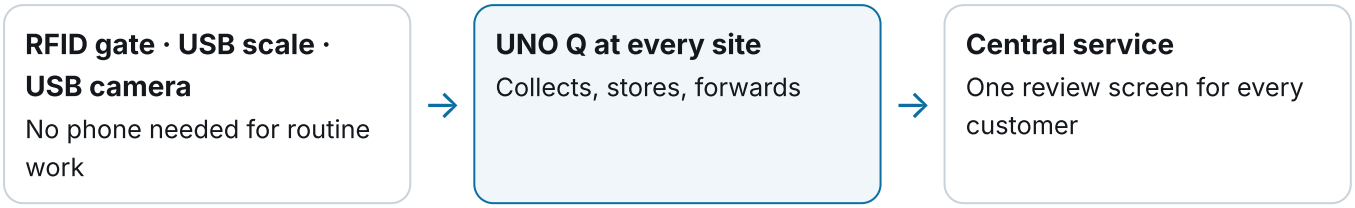
Short answer: one small, cheap board at each customer site collects every scan and weigh-in, saves it safely, and sends it on. It works without internet, keeps secrets off phones and browsers, and can later connect RFID readers, digital scales and USB cameras directly.

Low cost · Easy to maintain · Secure network · Easy to scale · Upgradeable

TODAY (DEMO)



LATER (PRODUCTION)



Q1 What does the board do today?

- Runs the ThreadTrace server and a small database (SQLite) on the board.
- Saves every scan, repair, write-off and weigh-in **first**, then sends it to the big screen.
- Sends the scale photo to Google Gemini to read the weight. The API key stays on the board.
- Shows feedback on the LED matrix: a check mark for a scan, a water level for a weigh-in.

Q2 Why not just use a phone?

A phone is good for scanning, but it is personal, it goes home with the worker, and the battery runs out. A gateway stays at the site, is always on, and keeps the data in one place. Any phone can use it, with no app to install.

Q3 Why not only the cloud?

A cloud server needs internet all the time, and it cannot plug into a scale or an RFID reader at the site. The gateway does the local work. In production, the gateway and the cloud work together: the gateway collects, the cloud shows the review for every customer.

Q4 Is it low cost?

- The board costs about **$59** (UNO Q 2GB, after the July 2026 price change). No laptop or mini PC is needed at the site.
- It is powered by USB-C, like a phone, so it can stay on all day with little energy.

- Workers use phones they already have. No special scanner is needed for the pilot.
- The software uses only free, open tools: Python, SQLite, plain web pages. No licence costs.
- Gemini is only called for a weigh-in photo, not for every scan, so AI costs stay small. With a USB scale (Q9), no AI call is needed at all.

Q5 Is it easy to maintain?

- **One folder, one command.** Update with `git pull` and restart with `run.sh`.
- **No extra Python packages** on the board. The server uses only the Python standard library, so there is little that can break.
- **One data file and one database file.** Easy to back up and easy to copy to a new board.
- **Remote support.** A technician can update the board over the private network, without a site visit.
- **Easy to replace.** If a board breaks, put in a new one and copy the folder. Events are stored with IDs, so nothing is counted twice.

Q6 Is it secure?

- **Private network.** The board is not open to the internet. Phones, board and screen talk over Tailscale, an encrypted private network (WireGuard).
- **Secrets stay on the board.** The Gemini API key is in a `.env` file on the board. It is never sent to a phone or a browser.
- **No personal data.** Events contain packages, garments, sites, weights and times. No names of wearers.
- **Data stays local first.** The customer site keeps its own copy, and only the needed events go to the central service.
- **Next steps for production:** HTTPS, one key per device, user login for the dashboard, and signed software updates.

Q7 Is it easy to scale?

- **One gateway per site.** Every event has a site ID and its own ID, so many gateways can report to one service.
- **Same software everywhere.** Every board runs the same folder. Only the site settings change.
- **Standard messages.** Today: simple HTTP. At scale: MQTT or an IoT hub, which are made for thousands of devices.
- **Same screens.** The review screen does not change when more sites are added.

Q8 Can it be upgraded?

Yes. The UNO Q has two parts: a **Linux computer** (for USB devices, network and AI) and a **microcontroller** (for sensors, lights and buttons). So we can add hardware without changing the idea.

Upgrade	What it gives
4GB version of the board	More memory for more sites, more data or local AI
UHF RFID reader	Garments and packages are read automatically when they pass a gate
Digital scale with USB or serial port	The exact weight, without a photo and without AI

USB camera	Automatic photos of the scale or of packages (see Q9)
Buttons, buzzer, sensors on the microcontroller	Simple feedback and input for workers, without a phone
Small local AI model	Read the scale on the board, even without internet

Q9 How would a USB camera and scale work in the future?

1. A USB camera hangs above the scale at the return station.
2. A worker puts a bag or package on the scale. The board sees the weight is stable.
3. The board takes a photo by itself and reads the weight (Gemini or a local model). If the scale has a USB port, the board reads the weight directly.
4. The camera can also read the package QR code in the same photo, so the weight is linked to the right package.
5. The LED shows a check mark, and the big screen shows the water saved.

Result: no phone, no typing, no counting. The worker only puts the bag on the scale.

Q10 What if the internet or the board stops?

- **No internet:** scans are still saved on the board, and the central service gets them later. Only the Gemini scale reading waits; the worker can type the weight.
- **No power:** the database is a file on the board, so saved events are still there after a restart.
- **Broken board:** replace it and copy the folder (Q5).

Q11 What is working now, and what is a plan?

Feature	Status
Server, database, live screen, LED feedback on the UNO Q	**Working in the demo**
Phone QR scan, package check-in, repair, write-off	**Working in the demo**
Gemini reads a photo of a scale	**Working in the demo**
Private network with Tailscale	**Working in the demo**
USB scale, USB camera, RFID reader	**Plan (the board supports USB)**
Many gateways to one central service (MQTT / IoT hub)	**Plan**
HTTPS, device keys, signed updates	**Plan for production**

Price source: news report of the Arduino UNO Q price change (2GB: $44 → $59, 4GB: $59 → $79, from 6 July 2026), published 26 June 2026. Prices can differ by country and shop.